

4.4.2 Electromagnetic waves

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) electromagnetic spectrum; properties of electromagnetic waves	
(b) orders of magnitude of wavelengths of the principal radiations from radio waves to gamma rays	
(c) plane polarised waves; polarisation of electromagnetic waves	Learners will be expected to know about polarising filters for light and metal grilles for microwaves in demonstrating polarisation. HSW9
(d) (i) refraction of light; refractive index; $n = \frac{c}{v}$; $n \sin \theta = \text{constant}$ at a boundary where θ is the angle to the normal	at the interface between medium 1 and medium 2, $n_1 \sin \theta_1 = n_2 \sin \theta_2$
(ii) techniques and procedures used to investigate refraction and total internal reflection of light using ray boxes, including transparent rectangular and semi-circular blocks	PAG6
(e) critical angle; $\sin C = \frac{1}{n}$; total internal reflection for light.	